



Guidelines

HYPOPHOSPHATAEMIA

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisational Wide	Medical, Nursing, Pharmacy

HIGH RISK MEDICATION

- Potassium dihydrogen phosphate ampoules contain concentrated potassium ions which may cause potentially fatal toxicity if administered as a large bolus dose
- Please ensure careful product selection and infusion bag(s) are mixed by inverting and shaking thoroughly prior to administration

Introduction

This guideline provides guidance to all Sir Charles Gairdner Hospital (SCGH) staff involved in the prescription, administration, supply and management of phosphate in patients with hypophosphataemia. This document supports and complies with related legislation, policy and other related documents [here](#).

1.0 General Information¹⁻¹⁰

Phosphate is a major intracellular anion that is involved in a number of physiological processes including energy production, acid-base buffering and maintaining optimal leucocyte function. About 80 to 85% of the body's phosphate is found in bone, with the remaining 15 to 20% predominantly being in muscle and soft tissue. The approximate daily dietary phosphate requirement is 25 to 45 mmol with about 80% being absorbed by the jejunum. Over 90% of phosphate excretion is performed via glomerular filtration with reabsorption occurring at the proximal tubules.

The incidence of hypophosphataemia in the general hospitalised population is approximately 2 to 3%, with the incidence increasing up to 30% for those admitted to the Intensive Care Unit (ICU). The diagnosis of hypophosphataemia is frequently made based on blood phosphate levels as mild-to-moderate hypophosphataemia is usually asymptomatic.

Patients who are at an increased risk of hypophosphataemia include those who are malnourished (cancer, alcoholism), receiving total parenteral nutrition (TPN) without adequate phosphate supplementation, recovering from diabetic ketoacidosis (DKA) and taking phosphate binders.

PathWest reference range for plasma inorganic phosphate is 0.75 to 1.50 mmol/L.

2.0 Hypophosphataemia Classification^{2,4,6-11}

- MILD = 0.73 to 0.80 mmol/L
- MODERATE = 0.50 to 0.72 mmol/L
- SEVERE = <0.50 mmol/L

3.0 Hypophosphataemia Symptoms^{1,2,4,6,7,9-11}

- Mild-to-moderate hypophosphataemia is often asymptomatic
- Symptoms rarely occur until phosphate levels fall below 0.64 mmol/L with serious symptoms not occurring until levels reach lower than 0.32 mmol/L.
- Neuromuscular: myopathy, metabolic encephalopathy, seizures
- Cardiorespiratory: respiratory muscular failure, left ventricular dysfunction, arrhythmia
- Haematological: haemolysis, thrombocytopenia, leucocyte dysfunction

4.0 Management Strategies

- It is important to diagnose and treat the underlying cause(s) of hypophosphataemia

Common causes of Hypophosphataemia ^{1,4,5,7-11}	
Internal redistribution	• Increased insulin secretion/insulin therapy • Recovery from malnutrition or DKA • Acute respiratory alkalosis • Hungry bone syndrome
Decreased intestinal absorption	• Inadequate dietary intake • Inhibition of phosphate absorption (i.e. antacids) • Steatorrhoea and chronic diarrhoea • Vitamin deficiency or resistance
Increased urinary excretion	• Primary and secondary hyperparathyroidism • Hereditary hypophosphataemic rickets • Oncogenic osteomalacia • Drugs: e.g. acetazolamide, tenofovir, diuretics

- Consider no supplementation for patients with mild-to-moderate severity with no risk factors for phosphate deficiency
- Deficiency risk factors – malnourishment, alcoholism, DKA, TPN
- Phosphate requirements will be dependent on cause and chronicity
- Correct hypocalcaemia (if present) prior to replacing phosphate

5.0 Presentation¹²

- Oral
 - Sodium phosphate monobasic (Phosphate-Phebra®) 500mg dispersible table
 - each 500mg contains 16.1mmol of phosphate (PO_4^{3-}), 20.4 mmol of sodium (Na^+) & 3.1mmol of potassium (K^+)
- Intravenous
 - Potassium dihydrogen phosphate [KH_2PO_4] injection ampoule (1.361g /10mL)

- each 10mL contains 10mmol of phosphate (PO_4^{3-}) & 10mmol of potassium (K^+)
- Sodium dihydrogen phosphate [NaH_2PO_4] injection ampoules (1.56g /10mL)
 - each 10mL contains 10mmol of phosphate (PO_4^{3-}) & 10mmol of sodium (Na^+)
- **Potassium Dihydrogen Phosphate amps are only to be stocked in critical care areas (ICU, General HDU) as per a WA Health Department Operational Directive.**

6.0 Drug Interactions

- Oral¹²
 - Calcium supplements = increased risk of ectopic calcification
 - Antacids = potentially reduce phosphate absorption
 - Colecalciferol / calcitriol = increases gastric absorption of phosphate

7.0 Phosphate Dosage & Administration

	Treatment	Comments
MILD 0.73 to 0.80mmol/L	Asymptomatic ORAL Monobasic Sodium Phosphate (Phosphate-Phebra®) 500mg Tablets Dose: 1 to 2 tablets BD (32.2-64.4mmol/24hrs) ^{4,6,7,10-12} or Milk ~15mmol per 480mL ^{4,6}	• Consider no supplementation^{4,6,7,11} • Monitor PO_4^{3-} level daily and adjust dose accordingly^{7,11} • Diarrhoea & GI upset are common treatment limiting side effects^{1,6,11} • Dissolve tablets in a third to half a glass water⁷ • Halve dose in renal impairment⁷ • Consider IV ONLY if patient cannot tolerate oral^{7,11}
	If symptomatic – see under SEVERE If no/minor Symptoms ORAL Monobasic Sodium Phosphate (Phosphate-Phebra®) 500mg Tablets Dose: 2 tablets BD or TDS (64.4-96.6mmol/24hrs) ^{4,6,7,11,12} IV Sodium Dihydrogen Phosphate ^{5-8,12} Dose: 10 to 20mmol ^{5,7,11} IVC/CVC: 10mmol in 100mL over at least 2hrs Potassium Dihydrogen Phosphate ^{5-8,13} Dose: 10 to 20mmol ^{5,7,11} IVC: 10mmol in 250mL over at least 2hrs CVC: 10mmol in 100mL over at least 2hrs *ENSURE THOROUGH MIXING*	• Oral route is preferred However, IV phosphate may be used as an adjunct to oral if large doses are required as this can cause diarrhoea & GI upset^{1,6,11} • Check PO_4^{3-} level after every 10 to 20mmol dose of IV phosphate and at least once daily for oral phosphate • Non-critical care areas use Sodium Phosphate unless hypernatraemia or hypokalaemia present • NOTE: For fluid restricted patients Sodium phosphate can go in 50mL via CVC route
MODERATE 0.50 to 0.72mmol/L		
SEVERE <0.50mmol/L	Symptomatic IV Sodium Dihydrogen Phosphate ^{5-8,13} Dose: 20 to 50mmol/day ^{1,5,7,8,11} IVC/CVC: 10mmol in at least 50mL over 1hr Potassium Dihydrogen Phosphate ^{5-8,13} IVC: 10mmol in 250mL over 1hr ^{5-8,13} CVC: 10mmol in at least 50mL over 1hr (see comments) ^{3,5,8} *ENSURE THOROUGH MIXING*	• Check PO_4^{3-} level every 6 to 12hrs during IV supplementation^{4,7,11} • Check Ca^{2+} level every 6 to 12hrs^{6,11} • Non-critical care areas use <u>Sodium</u> Phosphate unless hypernatraemia or hypokalaemia present • Cardiac monitoring maybe required if administering more than 10mmol/hr of Potassium Phosphate^{6,13} • NOTE: 45mmol of Phosphate in 100mL over 3hrs has been safely administered in critical ill patients³

8.0 Monitoring^{4,7,11-13}

- U&Es including phosphate (PO_4^{3-}), calcium (Ca^{2+}) & magnesium (Mg^{2+}) at least once a day
- Signs of hypocalcaemia – monitor for numbness/tingling of lips, feet &/or hands
- Blood pressure – potential for a fall in BP (especially following IV administration)

9.0 Adverse Effects

- Oral^{1,6,10,12}
 - Diarrhoea
 - Nausea & vomiting
 - Abdominal pain
 - Metabolic: hypocalcaemia, hyperkalaemia, hypernatraemia, hyperphosphataemia
- Intravenous^{3,7,10-12}
 - Metabolic: hypocalcaemia, hyperkalaemia, hypernatraemia, hyperphosphataemia, hypomagnesaemia
 - Arrhythmias
 - Renal insufficiency
 - Hypotension

10.0 Compatibility

- Refer to the [Australian Injectable Drugs Handbook](#) for information.

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising policy and standards

- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [SCGOPHCG High Risk Medication Policy](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [Department of Health WA Mandatory Standards for Intravenous Potassium](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)
- [Australian Commission on Safety and Quality in Healthcare. Recommendation for terminology, abbreviations and symbols used in medicines documentation, December 2016](#)
- NSQHS Standards: 1,4,6,8

Standards, procedures, guidelines:

- [SCGH Nursing Practice Guideline No.51 Medication Management](#)
- [SCGH Nursing Practice Guideline No.4 Intravenous Therapy](#)

References

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Authorisation & Version Control

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Standards Applicable	NSQHS Standards (Version 2): 			

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

BRIEF SUMMARY

DRUG	INDICATION	PRESENTATION	PREPARATION	DOSAGE	ADMINISTRATION	COMMENTS
Sodium Dihydrogen Phosphate IV (NaH_2PO_4)	Hypophosphataemia (moderate-to-severe)	clear vial with clear, colourless solution	Central line Dilute 10mmol in at least 50-100 mL sodium chloride 0.9%/glucose 5% Peripheral line Dilute 10mmol in at least 100mL sodium chloride 0.9%/glucose 5%	Moderate Hypophosphataemia 10-20mmol	Moderate Hypophosphataemia 2hr for every 10mmol of phosphate Severe Hypophosphataemia 1hr for every 10mmol of phosphate	- Non-critical care areas use Sodium Phosphate unless hypernatraemia or hypokalaemia present - Ensure thorough mixing after phosphate addition 10mmol/50mL solutions can be used via CVC in fluid restriction - Frequently recheck PO_4^{3-} levels - Cardiac monitoring maybe required if administering more than 10mmol/hr of Potassium Phosphate - More rapid phosphate supplementation has been safely administered in critical ill patients - Potassium phosphate ampoules are stock only in critical care areas - Oral supplementation is preferred route for mild-to-moderate hypophosphataemia - Diarrhoea may be dose limiting s/e for oral phosphate
Potassium Dihydrogen Phosphate IV (KH_2PO_4)	Hypophosphataemia (moderate-to-severe)	clear vial with clear, colourless solution	Central line Dilute 10mmol in at least 50-100mL sodium chloride 0.9%/glucose 5% Peripheral line Dilute 10mmol in at least 250mL sodium chloride 0.9%/glucose 5%	Severe Hypophosphataemia 20-50mmol		
Monobasic Sodium Phosphate 500mg Tabs (Phosphate-Phebra®)	Hypophosphataemia (mild-to-moderate)	White, flat circular tablet	Dissolve tablets in a third to half a glass water	Moderate Hypophosphataemia 1-2 tabs TWICE a day (32.2-64.4mmol/24hrs) Severe Hypophosphataemia 2 tabs TWO-THREE times a day (64.4-96.6mmol/24hrs)		